

Matrix Mult Designs - Basic

Inputs

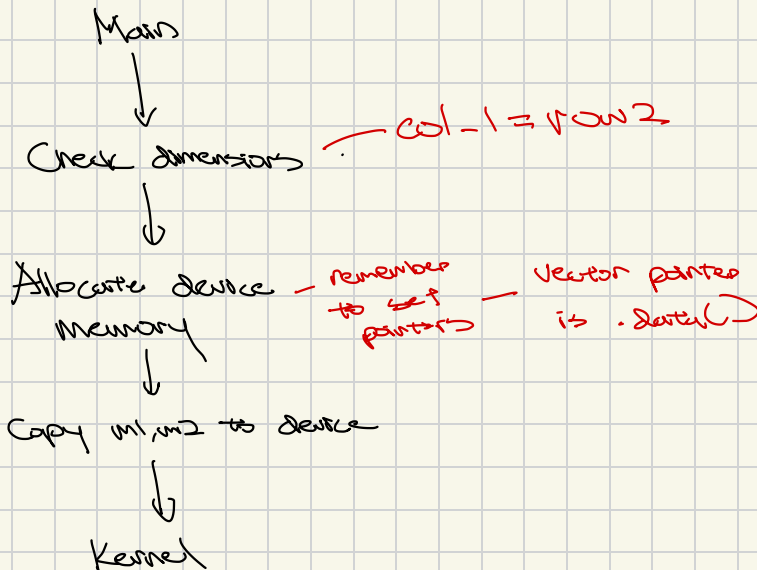
- Matrix $m1$ $[M \times K]$ vector
- Matrix $m2$ $[K \times N]$ vector
- Row of $m1$
- Col of $m2$ rest can be derived

Output

- Matrix $m3$ $[M \times N]$ vector

$$\begin{bmatrix} 0 & \dots & k \\ \vdots & & \vdots \\ m \end{bmatrix} \times \begin{bmatrix} 0 & \dots & N \\ \vdots & & \vdots \\ k \end{bmatrix} = \begin{bmatrix} e_0 & \dots & e_n \\ \vdots & & \vdots \\ e_m \end{bmatrix}$$
$$\begin{bmatrix} 0 \\ \vdots \\ m \end{bmatrix} \times [0 \dots N]$$

$m \times n$ threads total for each element



Matrix Mult kernel Simple

Inputs

- Pointers to $m1, m2, m3$
- Dimensions

Writes output element to $m3$

Each output elem $\Rightarrow$ thread

$$\begin{bmatrix} 0 & \dots & k \\ \vdots & & \vdots \\ m \end{bmatrix} \times \begin{bmatrix} 0 & \dots & N \\ \vdots & & \vdots \\ k \end{bmatrix} = \begin{bmatrix} e_0 & \dots & e_m \\ \vdots & & \vdots \\ e_n \end{bmatrix}$$

Diagram illustrating matrix multiplication. The first matrix has rows 0 to m and columns 0 to k . The second matrix has rows 0 to k and columns 0 to N . The result matrix has rows 0 to n and columns 0 to m . A green horizontal bar highlights row x in the first matrix and row x in the result matrix. A yellow vertical bar highlights column y in the second matrix and column y in the result matrix. The element e_i is highlighted in the result matrix at row x and column y .

Given some e_i in row x & col y

$$e_i = m1_{\text{row } x} \cdot m2_{\text{col } y}$$

$$\text{row} = i / \text{num cols}$$

$$\text{col} = i \% \text{num cols}$$

$$\begin{bmatrix} \uparrow & \uparrow \\ \text{start} & \text{++ indexing} \\ \text{row} \cdot M_1 & \text{num of cols} \end{bmatrix} \times \begin{bmatrix} \uparrow \\ \text{start col (row 0)} \\ \text{++ indexing} \\ \text{num cols of } m_2 \end{bmatrix}$$

Diagram illustrating the indexing for the matrix multiplication. The first matrix is indexed by row $\cdot M_1$ and num of cols (k). The second matrix is indexed by start col (row 0) and ++ indexing num cols of m_2 .

Should be 1 thread per result matrix element - max 1024 threads

Num blocks = $\frac{\text{elements} + (\text{num threads} - 1)}{\text{num threads}}$
 $\rightarrow$ learns better ways to synchronize mem across blocks

ceiling division